The research worker

Above: ARCHITECTURE.md §4.5.

Talks to:

- the desk, whose scheduler starts every run (JOBS.md);
- the data sources: SEC EDGAR, Form 13F, Polygon, Yahoo;
- the shared record, where it writes its facts;
- the tools' files, which it reads, read-only, to train models.

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1. What it is

The O'Neil research, done in one place. For every stock the tools find, and for the market as a whole, it works out what William O'Neil's method (CANSLIM) asks:

- how fast the company's earnings are growing (C and A);
- whether the stock leads: its strength against the whole market (RS), and its industry group's rank (L);
- whether it is breaking out of a sound base, or has something new (N);
- the supply of its shares (S), and whether big funds are buying (I);
- whether the market itself is in a confirmed uptrend (M).

It is a **job, not a program that stays up.** The desk's scheduler starts it; it works through its steps, writes each result with its date into the shared record, records its run, and **exits**.

It is the only writer of:

- every O'Neil fact: the letters above, and the ratings;
- the industry map: which industry each company belongs to, from its SEC filings (its SIC code);
- when a model is chosen, the tools' models.

What it never does:

- stay running;
- start on its own;
- make a page wait;
- write a tool's file.

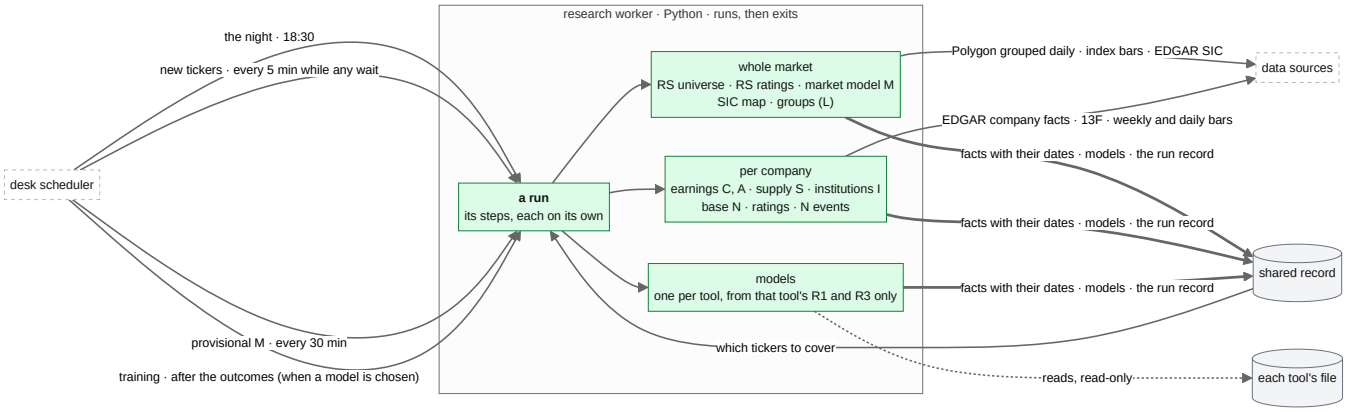
Why it is separate. Today the O'Neil research lives inside qp, the chart program, and each tool asks qp about each new stock:

- a stock first seen while qp is down or restarting gets a blank reading;
- a tool's morning record taken then keeps those blanks for good;
- five tools ask about the same stocks.

O'Neil results are card fields and Market-tab sections, so their writer must not depend on the chart program being up.

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2. The diagram



3. Its parts

3.1 The runs

Run	When (New York, trading days)	Steps
the night	18:30, once the day's whole-market bars are published	every step, then the backup (BACKUPS.md)
new tickers	every 5 minutes, 04:00–16:00, while any ticker carded today has no company facts and no attempt today	the per-company steps, for those tickers only
provisional M	every 30 minutes in the session	the market model on today's index bars so far, written as **provisional**; the night's run replaces it
training	after the outcomes (about 16:20), when a model is chosen	one model per tool, from that tool's own history only (LEARNING.md)
by hand	a button, or a command	any step. It asks the scheduler to start the job; no page waits

New tickers the same morning.

- EDGAR allows ten requests a second, so a morning's new tickers land within minutes.
- Until they land, the card says "computing · since 09:41".
- RS, the group and the SIC code already exist for every US stock, from the whole-market steps.

3.2 The steps

Each step is independent. One failure never stops the rest, and each failure is reported with its reason.

#	Step	Writes (fact family)	Clock
1	RS universe top-up	`bars_universe` (Polygon grouped daily, the newest sessions)	each close
2	RS ratings	`rs_rating` per ticker, with the universe size	each close
3	market model	`oneil_m`: state, per-index counts, distribution days, rally, follow-through, the rules in use	each close; every 30 minutes in the session, provisional

#	Step	Writes (fact family)	Clock
4	SIC map	`sic` per ticker	nightly, for new listings and changes
5	groups	`group_ranking`; `oneil_l` per ticker (group, level, rank, percentile, letter, RS in group, rotation)	each close
6	earnings	`oneil_c`, `oneil_a` per ticker	weekly, and on a new 10-Q or 10-K
7	supply	`oneil_s` per ticker	monthly (shares); each close (U/D, A/D)
8	institutions	`oneil_i` per ticker	quarterly, after the 45-day filing deadline
9	base and RS line	`oneil_n_base`, `rs_line` per ticker	each close (weekly bars)
10	ratings	`oneil_ratings`: RS, group, A/D, volume against its 50-day average; later EPS, SMR, Composite	each close
11	N events	`oneil_n_events` per ticker	nightly, over 24 months
12	distribution days per stock	`oneil_dd` per ticker	each close

Which tickers each night:

- **Whole-market steps (1–5)** cover every US stock.
- **Per-company steps (6–12)** cover:
    - every ticker any tool carded in the last 90 days;
    - every R1 ticker;
    - every ticker in an open trade.

  A ticker carded for the first time is covered the same morning by the new-ticker run.

Which tickers to cover is read from the shared record:

- the desk records each carded ticker (`research_wanted`);
- the worker records each attempt (`research_attempts`).

Budgets. The per-company steps have a time budget per night. Whatever does not fit is covered on the following nights, oldest first.

The method behind each letter, its rules, thresholds and traps: ONEIL.md.

3.3 Facts, with their dates

Every fact it writes carries:

- **`as_of`:** the date it describes (the quarter end, the session, the 13F quarter);
- **`known_at`:** when the worker learned it;
- **`status`:** `ok`, `none` (no filings) or `unavailable` (EDGAR timed out), with the reason;
- **the rules in use**, for anything with a threshold, so a signal can be checked against its definition later.

So a backtest's CANSLIM gate reads exactly the letters known before its decision bar, and a photo records exactly the reading of its minute.

3.4 Who reads what

Reader	Reads	Never
the desk's card pipeline	the O'Neil facts of the card's ticker (CARD.md)	computes any of them
the desk's market reader	the latest `oneil_m`, final or provisional, and `group_ranking` (MARKET.md)	computes any part of them
qp	any of them, as of a day (BACKTEST.md)	writes them
the journal	the facts known at a trade's entry (REVIEW.md)	writes them
pages	through the card and the market state	trigger a build

3.5 Its switch

`researchMode` (in `data/keys.json`) has three settings:

- **shadow:** the cards keep qp's values, and the facts are compared beside them field by field.
- **on:** the cards read the facts; qp is asked only for tickers not covered yet.
- **off.**

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4. Contracts

It owns (only writer), in the shared record	Subject
`bars_universe`, `group_ranking`, `oneil_m`	`MARKET`
`rs_rating`, `rs_line`, `sic`, `oneil_l`, `oneil_c`, `oneil_a`, `oneil_s`, `oneil_i`, `oneil_n_base`, `oneil_n_events`, `oneil_ratings`, `oneil_dd`	ticker
`research_attempts`, `research_runs`	—
models (when chosen)	tool, question, version

It reads	From
EDGAR (company facts, the ticker map, 8-K items, SIC), Form 13F datasets, Polygon grouped daily, Yahoo index and stock bars	the data sources (SOURCES.md)
the tickers to cover	`research_wanted`, written by the desk
R1 and R3 of a tool, to train its model	that tool's file, read-only

Its run record. Each run writes a `research_runs` row: its start and end, each step's result, and counts (tickers covered, facts written, failures with reasons). The scheduler's own run record holds the same outcome. A step that has not succeeded within its clock turns the health page amber, then red, and alerts. For example: earnings not refreshed in 8 days.

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5. When it fails

Event	What happens
a step fails	the other steps run. The failure is recorded with its reason, and the facts keep their old dates, so cards say "last run DATE"
a run does not happen, or hangs	the scheduler records it as missed or hung, and alerts
a source is slow or limited (EDGAR: ten requests a second)	the step works within its budget and carries the rest to the next night
it runs out of memory at night	the run fails and alerts. The night's run happens at 18:30, when no scan is due, with swap behind it (BOX.md)
the shared record is down	nothing can be written, so the run fails and alerts. Nothing is written anywhere else

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6. Built on

The code it runs: `quant-platform/research/worker.py`, reusing qp's O'Neil code, which was checked rule by rule: `relstrength.py`, `oneil.py`, `groups.py`, `edgar.py`, `f13.py`, `base.py`, `ratings.py`, `sic.py`.

The old desk:

- the O'Neil chain as built inside qp: legacy/qp/CANSLIM.md, 1,646 lines, the deepest document of the old desk;
- its nightly job: a systemd timer, Monday to Friday at 22:30 UTC, six steps (legacy/qp/OPERATIONS.md).

What failed there:

- The chain runs inside the chart program. Every tool asks qp for its new tickers' values and caches them for the day.
- A new ticker's earnings arrive only if a page shows its card, or the nightly walk reaches it.
- The nightly job runs from a hand-installed timer. Without it, the O'Neil data goes stale with no alarm (Q-N5).
- The industry map had two writers and flipped every morning (F-21).
- When an O'Neil file is missing, the Market tab asks qp to build it inside a page request.

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7. Status

Part	Status
the worker: the O'Neil readings written as dated facts; new tickers within minutes; provisional M	**built, in shadow** (step 7). The cards still read qp's values
its runs started by the desk	**partly**: the new-ticker and provisional-M runs are started by T1's scheduler; the night's run is the last step of qp's own nightly job, on its systemd timer
the whole chain inside the worker (steps 1–12), started by the desk's night job	**to build** (step C): today the chain's computations run in qp's nightly job and the worker publishes their results

Part	Status
`rs_line`, `oneil_s`, `oneil_n_events`, `sic` as facts; the ratings beyond RS, group and A/D	**to build** (ONEIL.md §6)
training models	**prepared, no model** (by the trader's choice; LEARNING.md)